

Demonstration of Proposed Features

Date	03 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the Smart Lender project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed ML-powered loan approval prediction solution.

S. No	Feature Name	Description	Status	Demonstrated (Yes/No)	Remarks
1	Data Preprocessing & Cleaning	Loaded applicant dataset, removed non-predictive Loan_ID column, and verified data quality	Implemented	Yes	Dataset (5000 records) had no missing values; Loan_ID dropped before training
2	Categorical Feature Encoding	Converted categorical fields (Gender, Married, Dependents, Education, Self_Employed, Property_Area, Loan_Status) into numeric form	Implemented	Yes	Used scikit-learn's LabelEncoder across all categorical columns
3	Multi-Model Comparison (Decision Tree, Random Forest, KNN, XGBoost)	Train and compare multiple classification algorithms to select the best-performing model	Partial	No	Only Random Forest implemented and evaluated so far; other algorithms pending
4	Model Hyperparameter Tuning	Tune Random Forest parameters to reduce overfitting and improve generalization	Implemented	Yes	Used GridSearchCV with 5-fold CV; testing accuracy improved from 68.7% to 70.6%
5	Model Persistence for Deployment	Save the best trained model for reuse in a prediction application	Implemented	Yes	Saved as loan_prediction_model.pkl using joblib
6	Flask Web App for Real-Time Prediction	Web interface for applicants/officers to submit details and instantly receive an approval prediction	Pending	No	Not yet integrated in the current notebook; planned for next phase

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	4
Total Features Demonstrated	4
Overall Implementation Rate (%)	66.7%